Tubular & Tubulointerstitial Diseases — I

Pathology GUE Block · MEDucation by Mattin Majid

This lecture's own classification: **Primary tubular disease** (ATN) vs **tubulointerstitial disease** (pyelonephritis) — the whole SGL follows this split, continued in Part II with renal stones and cystic kidney disease.

BEFORE YOU READ — ANSWER FROM MEMORY

1. What is the single urine finding that localises infection to the kidney rather than the lower urinary tract — and why does it do that?
2. Name the two forms of chronic pyelonephritis and what distinguishes them.
3. Which organ process is reversible — acute tubular necrosis or chronic pyelonephritis?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

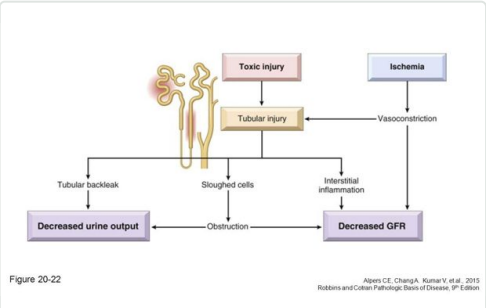
1 Classification

Group	Definition	Example
I. Primary tubular diseases	Tubular injury by ischaemic or toxic agents	Acute tubular necrosis
II. Tubulointerstitial diseases	Inflammatory involvement of the tubules and interstitium	Pyelonephritis (acute and chronic)

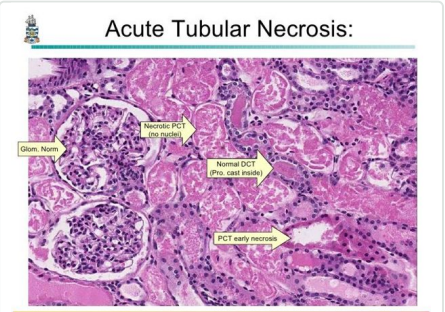
2 Acute Tubular Injury / Necrosis (ATN / ATI)

Significance	**Most common cause of acute kidney injury**; it is a **reversible** process.
Causes	**Ischemia:** PAN, malignant hypertension, hypovolemic shock. **Direct toxic injury:** drugs (**gentamicin**), heavy metals (mercury), myoglobin (skeletal muscle injury), radiation. **Acute tubulointerstitial nephritis** due to drug hypersensitivity (not bacterial). Disseminated Intravascular Coagulation (DIC). **Urinary obstruction:** tumors, BPH, or blood clots (post-renal acute renal failure).
Pathogenesis	Toxic injury or ischemia (via vasoconstriction) → **tubular injury** → tubular backleak + sloughed cells + interstitial inflammation → decreased urine output and decreased GFR (sloughed cells also cause obstruction).
What you see	Attenuation of tubule luminal cells with increased eosinophilic staining; tubule lumens filled with **hyaline and fibrin**; necrotic PCT (no nuclei) beside normal DCT.

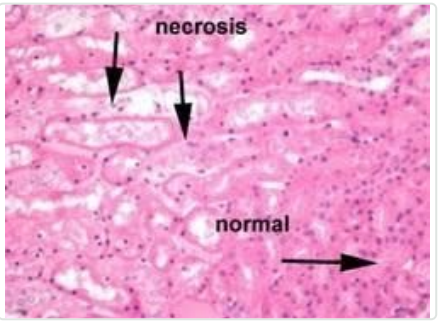
Source note: the lecture lists "acute tubulointerstitial nephritis due to hypersensitivity" as one of the *causes* of ATN. Strictly this is a distinct disease process (an interstitial, not purely tubular, injury) rather than a mechanism of necrosis — kept here exactly as the lecture states it.



Pathogenesis: toxic injury / ischemia → tubular injury → decreased urine output / decreased GFR



ATN histology — necrotic PCT (no nuclei) vs normal DCT (protein cast inside)



Necrosis vs normal tubule, side by side

3 Tubulointerstitial Nephritis (TIN) — overview

Renal diseases involving **inflammatory injuries of the tubule & interstitium**, often insidious in onset, principally manifest by **azotemia**. Two forms: **acute** and **chronic** TIN.

Category	Causes
Infections	Acute bacterial pyelonephritis; chronic pyelonephritis (incl. reflux nephropathy); other infections (viruses, parasites)
Toxins	Drugs; acute hypersensitivity interstitial nephritis; analgesics; heavy metals (lead, cadmium)
Metabolic diseases	Urate nephropathy; nephrocalcinosis (hypercalcemic nephropathy); oxalate nephropathy
Physical factors	Chronic urinary tract obstruction; neoplasms; multiple myeloma (light-chain cast nephropathy)
Immunologic reactions	Transplant rejection; Sjögren syndrome; sarcoidosis
Vascular diseases	(listed as a category; no specific entities given in this lecture)
Miscellaneous	Nephronophthisis; "idiopathic" interstitial nephritis

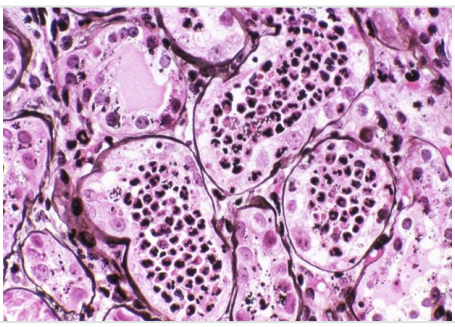
4 Pyelonephritis

One of the **most common diseases of the kidney** — inflammation affecting the tubules, interstitium, and **renal pelvis**. Two forms: acute and chronic.

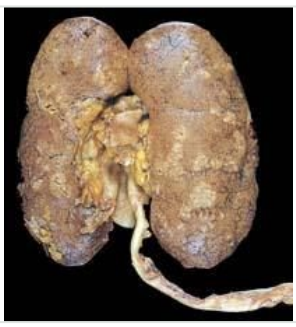
Cause	Detail
Commonest (85%)	Gram-negative bacteria, normal intestinal flora — **E. coli**, Proteus, Klebsiella, TB…
Immunocompromised	Polyomavirus, CMV…

Acute pyelonephritis — illness script

Route to kidney	**Hematogenous**, or **ascending** (via vesicoureteral reflux).
Presentation	Sudden onset — pain, fever, malaise. Bladder/urethral irritation: **dysuria, frequency, urgency**.
Key discriminator	Urine contains many pus cells — **pyuria** — but pyuria alone does **not** differentiate upper from lower UTI. **Leukocyte casts (pus casts)** do, because casts form only in the tubules.
Diagnosis	Established by **urine culture**.
What you see	Gross: cortical surface shows grayish-white areas of inflammation and abscess formation. Microscope: patchy interstitial suppurative inflammation, intratubular aggregates of neutrophils, and tubular necrosis.
Complications	**1. Papillary necrosis** — mainly diabetics, and patients with urinary tract obstruction. **2. Pyonephrosis** — with total obstruction, suppurative exudate cannot drain and fills calyces, pelvis, and ureter. **3. Perinephric abscess** — suppurative inflammation extends through the renal capsule into perinephric tissue.



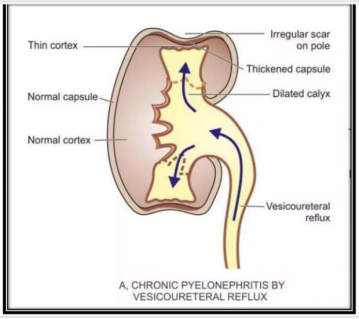
Acute pyelonephritis — intratubular neutrophil aggregates



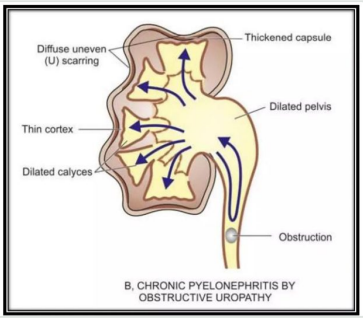
Acute pyelonephritis — grayish-white cortical abscesses

Chronic pyelonephritis

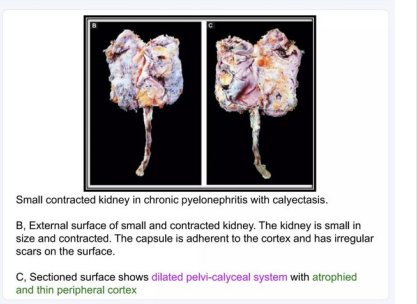
Definition	Chronic tubulointerstitial inflammation and **renal scarring** associated with involvement of the renal calyces and pelvis.
Two forms	**Chronic reflux-associated** pyelonephritis (congenital vesicoureteral reflux) and **chronic obstructive** pyelonephritis (renal stone).
Gross	Kidneys irregularly scarred. On cut section: coarse, discrete, **corticomedullary scar overlying a dilated, blunted, or deformed calyx**. Small, asymmetrically contracted kidney with depressed (U-shaped) scars, blunted calyces, dilated pelvis.
Microscope (M/F)	Tubules show **atrophy** in some areas, **dilation** in others. Tubules lined with flattened epithelium may be filled with colloid casts (**thyroidisation**). Chronic interstitial inflammation and fibrosis. Glomeruli may appear normal except for **periglomerular fibrosis**.



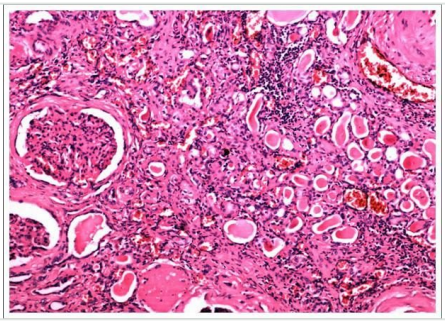
Chronic pyelonephritis by vesicoureteral reflux



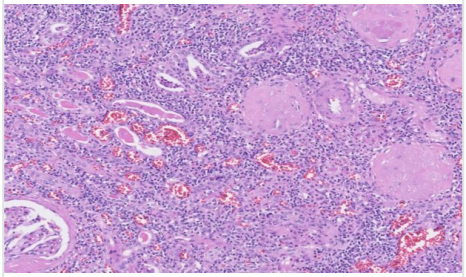
Chronic pyelonephritis by obstructive uropathy



Small, irregularly contracted kidney with calyectasis



Thyroidisation — colloid-filled tubules



Chronic interstitial inflammation and fibrosis

Acute vs chronic pyelonephritis — the discriminator

Feature	Acute	Chronic
Core process	Suppurative (neutrophilic) inflammation, abscess formation	Scarring with chronic interstitial inflammation + fibrosis
Calyces/pelvis	Not structurally deformed	**Deformed, dilated, or blunted** — defines the diagnosis
Tubules	Intratubular neutrophils, tubular necrosis	Atrophy/dilation, thyroidisation (colloid casts)
Mechanism	Ascending or hematogenous infection	Reflux *or* obstruction (renal stone) sustaining recurrent/ persistent infection

P Must-remember pearls

- Pyuria does not localise infection — **leukocyte (pus) casts** do, because casts form only inside tubules.
- ATN is the **most common cause of AKI** and is **reversible**; chronic pyelonephritis causes permanent scarring.
- Chronic pyelonephritis is defined by **calyceal/pelvic scarring** — that is what separates it from other causes of chronic TIN.
- Two routes to chronic pyelonephritis, same endpoint: **reflux** (congenital vesicoureteral reflux) or **obstruction** (renal stone).
- 85% of pyelonephritis is caused by normal gut Gram-negative flora, with **E. coli** as the prototype.

Q Test yourself

1. A patient develops acute kidney injury after a crush injury with extensive skeletal muscle damage. Which mechanism of ATN does this represent?

A. Ischemic injury
B. Direct toxic injury (myoglobin)
C. Urinary obstruction
D. Hypersensitivity interstitial nephritis

2. Two patients have pyuria on urinalysis. Only one also has leukocyte casts. This additional finding tells you that:

A. Both infections are equally likely to be upper tract
B. The patient with casts has a lower UTI only
C. The patient with casts has renal (tubular) involvement
D. Casts rule out pyelonephritis

3. A child with known congenital vesicoureteral reflux develops progressive renal scarring over several years. Which form of chronic pyelonephritis is this?

A. Chronic obstructive pyelonephritis
B. Chronic reflux-associated pyelonephritis
C. Acute pyelonephritis
D. Papillary necrosis

4. A diabetic patient with acute pyelonephritis and underlying urinary tract obstruction develops sudden flank pain and sloughed renal tissue in the urine. The most likely complication is:

A. Perinephric abscess
B. Pyonephrosis
C. Papillary necrosis
D. Chronic obstructive pyelonephritis

5. Which of the following is an immunologic cause of tubulointerstitial nephritis, per the lecture's classification?

A. Multiple myeloma
B. Sjögren syndrome
C. Lead exposure
D. Chronic urinary tract obstruction

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 · B — Myoglobin released from damaged skeletal muscle is a direct tubular toxin.

2 · C — Leukocyte casts form only in the tubules, so their presence localises the infection to the kidney.

3 · B — Congenital vesicoureteral reflux defines chronic reflux-associated pyelonephritis.

4 · C — Papillary necrosis is linked to diabetics and patients with urinary tract obstruction.

5 · B — Sjögren syndrome is listed under immunologic reactions, alongside transplant rejection and sarcoidosis.

SPACED REVIEW

☐ Day 1	☐ Day 3	☐ Day 7	☐ Day 21	☐ Pre-exam